WORKED SOLUTION

STEP I 1989, Question 16

Probability and Statistics • Zero-Sum Games

STEP I 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.14, *Mixed Strategies*: https://beyondstep.keithmcnulty.org/ch12/mixed-strategies.html

The key idea

Everything follows from one small table: B's winnings for each of the nine pairs of guesses. Once that is written down, each expected value is just a weighted sum of table entries. For the last part, grouping the sum by A's probabilities a_j turns "whatever A does" into a condition on three coefficients, and making all three non-negative is a small system of linear inequalities.

Solution

Write $w(i,j)$ for B's winnings (in pounds) in a round where A guesses i and B guesses j. Reading off the rules:

- if $|i-j|=2$, the one who said 1 pays the one who said 3 two pounds, so $w(1,3)=2$ and $w(3,1)=-2$;
- if $|i-j|=1$, the one who said the larger number pays one pound, so $w(2,1)=w(3,2)=1$ and $w(1,2)=w(2,3)=-1$;
- $w(i,i)=0$.

	B says 1	B says 2	B says 3
A says 1	0	-1	2
A says 2	1	0	-1
A says 3	-2	1	0

Notice that $w(j,i)=-w(i,j)$: the rules treat the players identically, so swapping the guesses swaps who pays.

Part (i)

If A always says N, B should look along row N of the table and always pick the largest entry.

- $N = 1$: B always says 3 and wins £2 every round.
- $N = 2$: B always says 1 and wins £1 every round.
- $N = 3$: B always says 2 and wins £1 every round.

In each case this choice is the unique best one, and every other reply wins nothing or loses.

Part (ii)

In a given round A's guess and B's guess are independent (A randomises on his own, and B decides secretly), so the probability that A says i and B says j is $\frac{1}{3}b_j$. Hence B's expected winnings are

$$\begin{aligned} \mathrm{E}(W) &= \sum_{j=1}^{3} b_j \cdot \tfrac{1}{3}\big(w(1,j) + w(2,j) + w(3,j)\big) \\ &= \tfrac{1}{3}\big(b_1(0+1-2) + b_2(-1+0+1) + b_3(2-1+0)\big) \\ &= \tfrac{1}{3}\,(b_3 - b_1). \end{aligned}$$

Since $b_3 \le 1$ and $b_1 \ge 0$, this is at most $\frac{1}{3}$, with equality exactly when $b_3 = 1$ (which forces $b_1 = b_2 = 0$). So B should *always guess* 3, and then expects to win £$\frac{1}{3}$ per round. Against a uniformly random A, the guess 3 wins £2, loses £1 or draws, each with probability $\frac{1}{3}$.

Part (iii)

With A guessing i with probability a_i and B guessing j with probability b_j, independently,

$$\mathrm{E}(W) = \sum_{i=1}^{3}\sum_{j=1}^{3} a_i b_j\, w(i,j).$$

Collecting the terms for each a_i (one row of the table at a time):

$$\mathrm{E}(W) = \underbrace{(2b_3 - b_2)}_{X}\, a_1 + \underbrace{(b_1 - b_3)}_{Y}\, a_2 + \underbrace{(b_2 - 2b_1)}_{Z}\, a_3.$$

We want $X \ge 0$, $Y \ge 0$ and $Z \ge 0$, that is

$$b_2 \le 2b_3, \qquad b_3 \le b_1, \qquad 2b_1 \le b_2.$$

Chaining these, $b_2 \le 2b_3 \le 2b_1 \le b_2$, so all of them must be equalities: $b_3 = b_1$ and $b_2 = 2b_1$. Together with $b_1 + b_2 + b_3 = 1$ this gives

$$b_1 = \tfrac{1}{4}, \qquad b_2 = \tfrac{1}{2}, \qquad b_3 = \tfrac{1}{4},$$

and then $X = Y = Z = 0$, which are certainly non-negative. So such a choice exists (and in fact it is the only one).

Ask yourself Could B do better and make X, Y and Z all strictly positive?

No. For any b_j,

$$b_1X + b_2Y + b_3Z = b_1(2b_3 - b_2) + b_2(b_1 - b_3) + b_3(b_2 - 2b_1) = 0,$$

which is just the statement that B playing against a copy of himself breaks even (the table is antisymmetric). A weighted sum of X, Y, Z with non-negative weights summing to 1 can't be zero if all three are positive. That is why the inequalities above could only be satisfied with equality.

Now the deduction. With $b = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ we have

$$\mathrm{E}(W) = 0 \cdot a_1 + 0 \cdot a_2 + 0 \cdot a_3 = 0$$

for *every* choice of a_1, a_2, a_3; more generally, whenever $X, Y, Z \geq 0$, $\mathrm{E}(W) = Xa_1 + Ya_2 + Za_3 \geq 0$ because each $a_i \geq 0$. B's winnings on different rounds are independent and identically distributed (both players' guesses are independent from round to round), each with mean at least 0. By the law of large numbers, B's average winnings per round over many rounds tend to this mean, so in the long run B loses no money, whatever A does. □

Checks

- Part (ii) is the special case $a_1 = a_2 = a_3 = \frac{1}{3}$ of part (iii): $\frac{1}{3}(X + Y + Z) = \frac{1}{3}(b_3 - b_1)$, as found.
- Part (i) is the case where a puts all its weight on N: for $N = 1$, $\mathrm{E}(W) = X = 2b_3 - b_2$, maximised by $b_3 = 1$, giving 2.
- A simulation of 2×10^5 rounds with B using $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ against A using $(0.6, 0.1, 0.3)$ gave average winnings of about -0.004 per round, consistent with 0.

Common slips

- Getting the sign convention backwards: in a difference of 1 it is the *larger* guess that pays, but in a difference of 2 it is the *smaller*. Build the table first and check it is antisymmetric.
- Concluding from part (ii) that B should mix his guesses. Against a fixed mixed strategy, the best reply is always a pure one (here, always 3); mixing only pays when B must guard against *every* strategy of A, as in part (iii).

Beyond STEP

This is a two-player zero-sum game, and part (iii) is a small case of von Neumann's minimax theorem: B has a mixed strategy guaranteeing him at least the value of the game whatever A does, and because the game is symmetric that value is 0. Finding such strategies is a linear programming problem, which connects probability with optimisation and economics. For the probability side, see Grimmett and Welsh, *Probability: An Introduction*; for games themselves, a good first read is Binmore, *Playing for Real*.

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